A fixed smooth metric violating Yau's nodal upper bound

An exposition of OpenAI's construction

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Abstract

We present the construction of OpenAI's September 2026 preprint *Power-law violations of Yau's nodal upper bound*. It produces one smooth Riemannian metric on $S^4 \times S^1$ and a sequence of real exact eigenfunctions whose intrinsic nodal four-volume exceeds the square root of the eigenvalue by a fixed positive power. The account follows the local envelope and packet constructions through exact coefficient correction, spectral persistence, and the smooth limit. Particular attention is given to the common exponent and to the order in which frequencies and perturbation tolerances are chosen. The counterexample and its mathematical mechanism are due to OpenAI; this note is an attributed exposition and literature update.

1 The question and the result

The smooth upper-bound question asks whether, on every closed connected smooth Riemannian manifold (M, g) of dimension $d \geq 2$, there is a finite constant $C(M, g)$ such that

$$\mathcal{H}_g^{d-1}(\mathcal{Z}(u)) \leq C(M, g)\sqrt{\lambda}, \qquad -\Delta_g u = \lambda u, \quad \lambda > 0, \tag{1.1}$$

for every nonzero real eigenfunction. Here $\mathcal{Z}(u) = \{x : u(x) = 0\}$, and Hausdorff measure is taken for the distance of the same metric g. The constant must be uniform over eigenvalues and eigenfunctions. This is the formulation of AIM Problem 011 at the revision recorded in [2].

OpenAI's preprint [1, Theorem 1.1] gives the following counterexample. All substantive construction below follows that source; the exposition does not claim a new counterexample.

Theorem 1.1 (OpenAI)**.** *There are a smooth positive Riemannian metric g on $S^4 \times S^1$, a number $\epsilon_0 > 0$, and nonzero real smooth functions u_j satisfying*

$$-\Delta_g u_j = \lambda_j u_j, \qquad \lambda_j > 0, \qquad \lambda_j \longrightarrow \infty,$$

such that

$$\frac{\mathcal{H}_g^4(\mathcal{Z}(u_j))}{\lambda_j^{1/2+\epsilon_0}} \longrightarrow \infty. \tag{1.2}$$

One admissible example in dimension five negates the universal assertion (1.1). Indeed, (1.2) makes the ratios $\mathcal{H}_g^4(\mathcal{Z}(u_j))/\sqrt{\lambda_j}$ unbounded for this fixed metric. This deduction uses the quantifier over manifolds in the question; it does not assert a separate result in every dimension.

The construction first works with a weighted elliptic operator on S^4, whose tensor and density can be perturbed independently. Local complex waves create many pairs of disks on which a real

function has opposite signs. An explicit correction turns this function into an exact eigenfunction with arbitrarily small changes of the coefficients. A protected round region then permits a further perturbation that makes its eigenvalue simple while retaining the function. Simplicity and strict signs allow every earlier construction to survive later changes. Summable perturbations in every finite derivative norm produce one smooth limiting operator. A warped circle realizes that operator as the ordinary Laplacian of a fixed five-dimensional metric.

The quantitative point is that the local gain has one exponent $\gamma > 0$, independent of the finite accuracy subsequently requested. At stage j, the frequency can therefore absorb its stage-dependent certificate constant. Sections 3–5 prove the local construction. Section 6 supplies the single metric and the intrinsic measure estimate.

2 The weighted operator and a quantitative step

Let $B = S^4 \subset \mathbb{R}^5$, with round metric g_0 and round density μ. Write points as (x_1, x_2, y), $y \in \mathbb{R}^3$, and put $t = |y|^2$. A positive pair $\mathcal{P} = (A, \rho)$ consists of a smooth positive definite symmetric contravariant tensor A and a smooth positive function ρ. Its principal metric and weighted operator are

$$G^{-1} = A/\rho, \qquad Lf = \rho^{-1} \operatorname{div}_\mu(A\,\mathrm{d}f). \tag{2.1}$$

Thus $-L$ is nonnegative and symmetric in $L^2(\rho\,\mathrm{d}\mu)$, with energy $\int_B A(\mathrm{d}f, \mathrm{d}f)\,\mathrm{d}\mu$ and form domain $H^1(B)$. The shifted energy obtained by adding $\int_B \rho f^2\,\mathrm{d}\mu$ is coercive.

Choose a finite smooth atlas with compact subpatches covering B. The C^r norms include all coordinate derivatives through order r, and the pair norm is the sum of its component norms. We use monotone norms in r. All pairs stay in one small C^1 neighborhood of $(g_0^{-1}, 1)$; hence their metrics, inverse metrics and connections have common bounds and common ellipticity constants. Higher derivative bounds need only be finite for a fixed input pair.

Envelope gradients and Hessians refer to G. For a real function h define

$$p = \nabla_G h, \quad Q = (1 + |p|_G^2)^{1/2}, \quad J_h(e) = \nabla_G^2 h(p/Q, p/Q) + \nabla_G^2 h(e, e), \quad |e|_G = 1, \quad \mathrm{d}h(e) = 0. \tag{2.2}$$

At $p = 0$ every unit vector is admissible. Complex tensor evaluations are bilinear; norms of complex vectors and matrices are Hermitian. Fix $\delta = 10^{-6}$. Constants c, C, C_r can change between occurrences; their important dependencies are specified locally.

For an integer $N > 1$, set

$$k = \sqrt{N(N+3)}, \qquad u_* = \operatorname{Re}(x_1 + ix_2)^N. \tag{2.3}$$

The polynomial is homogeneous and harmonic in $\mathbb{R}^5$. Substitution into $\Delta_{\mathbb{R}^5} = \partial_r^2 + 4r^{-1}\partial_r + r^{-2}\Delta_{g_0}$ gives $-\Delta_{g_0} u_* = k^2 u_*$. Where $t < 1$,

$$u_* = (1-t)^{N/2} \cos(N\vartheta),$$

with ϑ the angle in the (x_1, x_2)-plane. Part of a round region will retain this exact solution.

Definition 2.1. In a fixed coordinate chart $\Omega \subset \{t > 1/2\}$, a sign certificate for a continuous real function is a finite collection of pairs of congruent closed parallel three-disks D_a^-, D_a^+. Their joining closed cylinders lie compactly inside pairwise disjoint open coordinate balls. The function has strictly opposite signs on the two disks of each pair. The certificate size is

$$|\mathcal{C}| = \sum_a \mathcal{H}^3_{\mathrm{Euc}}(D_a^+).$$

Every joining segment contains a zero. Strict signs on finitely many compact disks persist under a sufficiently small C^0 change, and under multiplication by any nonzero real scalar.

Proposition 2.2 (OpenAI's quantitative step)**.** *There are a C^1 neighborhood $\mathcal{U}$ of the round pair, a chart Ω as above, and a number $\gamma > 0$, with this property. Fix a smooth $(A, \rho) \in \mathcal{U}$ that is round on $\{t < l_+\}$, where*

$$1/4 \le l_- < l_+ \le 1/2.$$

For every integer $r \ge 0$ and every $\eta > 0$ there are $c > 0$ and N_0 such that, for every integer $N \ge N_0$, there are a smooth positive pair $(\widehat{A}, \widehat{\rho}) \in \mathcal{U}$ and a nonzero real smooth function u for which

$$\|(\widehat{A} - A, \widehat{\rho} - \rho)\|_{C^r} < \eta, \qquad (\widehat{A}, \widehat{\rho}) = (g_0^{-1}, 1) \quad (t < l_-), \tag{2.4}$$

$$-\widehat{L}u = k^2 u, \qquad k^2 \text{ is simple on } B, \tag{2.5}$$

$$|\mathcal{C}| \ge c k^{1+\gamma} \qquad \text{for a sign certificate in } \Omega. \tag{2.6}$$

Here k is given by (2.3)*. The neighborhood, chart and exponent are common to all steps. The constant c and threshold N_0 may depend on the fixed pair, collar, norm order and tolerance.*

The next three sections give the proof, following [1, Proposition 2.2 and Sections 3–5]. The location of c before the universal frequency quantifier in this statement will be essential in Section 6.

3 A common power gain in the envelope

We construct $h = h_k$ with a positive test (2.2) wherever packets will be centered, while making the integral of its local oscillation scale grow by a power of k.

Proposition 3.1. *There are a small C^1 neighborhood $\mathcal{G}$ of g_0, fixed regions $V \Subset \Omega \subset \{t > 1/2\}$, and common $a_*, c, \gamma > 0$, such that every fixed smooth $G \in \mathcal{G}$ admits envelopes h_k satisfying*

$$\begin{aligned} h_k &= -a_* t + c_k \quad (t \le 1/2), & \|h_k\|_{C^j} &\le C_j k^{\delta(j+1)} \quad (j \ge 0), \\ \max_e J_{h_k}(e) &\ge c k^{-\delta} \quad (t \ge 1/4), & \int_V (1 + |\nabla_G h_k|_G^2)^{1/2} \, \mathrm{d}x &\ge c k^{\gamma}. \end{aligned} \tag{3.1}$$

The additive constants c_k may be any prescribed bounded sequence. For each fixed derivative order the constant and frequency threshold can depend on that order and on the fixed smooth metric. The exponent γ is independent of these orders, all later residual accuracies, and all round gluing collars.

3.1 A seed and a repeated local modification

The round seed $\widetilde{h} = -a_* t$ obeys

$$|\nabla_{g_0} \widetilde{h}|^2 = 4 a_*^2 t(1 - t), \qquad \nabla_{g_0}^2 \widetilde{h} = 2 a_* (t g_0 - \mathrm{d}y \cdot \mathrm{d}y).$$

Since $\mathrm{d}y : T_x B \to \mathbb{R}^3$ has a four-dimensional domain, there is a unit vector in its kernel. It is an admissible test, with Hessian value $2a_* t$. The Hessian norm is at most $2a_*$, so for $t \ge 1/4$ and $a_*^2 \le 1/8$,

$$J_{\widetilde{h}}(e) \ge a_*/2 - 2a_*^3 \ge a_*/4.$$

At a point with $t = 1$, the Hessian has two eigenvalues $2a_*$ and two zero eigenvalues. Add a sufficiently small smooth quadratic bump there, supported in $\{t > 1/2\}$, to make the Hessian positive definite at that point. Throughout the bump support two positive directions remain and the gradient stays sufficiently small. Their two-plane intersects $\ker \mathrm{d}h$, leaving a positive test even at a critical point. Away from that support the old kernel test applies. Choose a nearby region U_0 where the Hessian is positive definite and the gradient is separated from zero. These properties persist in a common small C^1 metric neighborhood: the Hessian of a fixed function changes by the connection difference contracted with its differential. Fix $U_0 \Subset V \Subset \Omega$.

A cube P of side νS, $S \le 1$, is called prepared for h if, on its closure,

$$\begin{gathered} |\nabla h| > 0, \qquad |\nabla^2 h| \le C_{\mathrm{p}} |\nabla h|/S, \\ \text{there is a two-plane } E \subset \ker \mathrm{d}h \text{ with } J_h(e) \ge c_{\mathrm{p}} |\nabla h|/S \text{ for every unit } e \in E. \end{gathered} \tag{3.2}$$

The plane need not depend continuously on the point. The seed is prepared at scale 1 on a sufficiently small cube in U_0, for fixed large C_{p} and small $c_{\mathrm{p}} > 0$.

Lemma 3.2. *One can fix a reciprocal integer $\nu > 0$, constants $0 < \alpha \le 1 \le \beta$, and $\zeta > 0$, uniformly under the common metric bounds, so that every prepared parent P admits a smooth modification $v = h + w$, supported in its interior, with*

$$\alpha |\nabla h| \le |\nabla v| \le \beta |\nabla h|, \qquad \max_e J_v(e) \ge c |\nabla h|/S.$$

Put $\ell = \nu^2 S$. Some cubes of side $\nu\ell$ inside P are prepared at scale ℓ, with the same constants, and their union $\mathcal{R}_P$ has

$$\int_{\mathcal{R}_P} |\nabla v| \, \mathrm{d}x \ge (1 + \zeta) \int_P |\nabla h| \, \mathrm{d}x. \tag{3.3}$$

If $s = |\nabla h(x_P)|$ at the parent center, then

$$\|D^j w\|_\infty \le C_j s \ell^{1-j} \qquad (j \ge 0). \tag{3.4}$$

Neither the modification nor these bounds use higher derivatives of the input h.

Proof. Choose a smooth $F : [0, 1] \to \mathbb{R}$, positive inside, equal to its argument near zero and flat at one. In a finite collection of disjoint balls of radii R_i in a unit three-cube, define

$$\psi(r) = -\int_r^{R_i} F(q/R_i) \, \mathrm{d}q, \qquad f = F(r/R_i), \qquad u_0 = f/r.$$

Extend by zero and periodically. At the center ψ is a constant plus $r^2/(2R_i)$, so it is smooth there; flatness makes the boundary extension smooth. With radial unit vector σ,

$$D\psi = f\sigma, \qquad D^2\psi = f'\sigma \otimes \sigma + u_0(I - \sigma \otimes \sigma). \tag{3.5}$$

The inequalities $f^2|f'| \le Cu_0$ and $f^2 \le Cu_0$ hold. The first constant does not depend on a ball radius, since $f^2|f'|/u_0 = vF(v)|F'(v)|$, $v = r/R_i$. Also $u_0 = 0$ implies $f = f' = \psi = 0$. At a center the radial Hessian is independent of the choice of σ.

Fix a small amplitude $a > 0$ using that radius-independent bound. The mean gradient factor inside any ball is

$$m(a) = 3 \int_0^1 v^2 \sqrt{1 + a^2 F(v)^2} \, \mathrm{d}v > 1.$$

Let $\eta_0 = (m(a)-1)/8$. Choose the finite packing to have volume fraction ϖ with $\varpi m(a) > 1+6\eta_0$. Such finite packings can approach full volume: finitely many grid cubes capture a fixed fraction of any remaining open volume; smaller inscribed balls capture a fixed fraction of those cubes. Repeating this finitely often leaves an arbitrarily small uncovered volume and keeps closures disjoint. Reduce each ball to a concentric core of relative radius $1-\eta_{\mathrm{c}}$, so that the integral per period over the cores is $C_{\mathrm{core}} > 1+5\eta_0$. Choose an inner-parent volume fraction $q_{\mathrm{in}} < 1$ with $q_{\mathrm{in}} C_{\mathrm{core}} > 1+4\eta_0$. A smooth parent cutoff χ, equal to one on that inner cube and flat on its zero set, is now fixed.

Take affine coordinates $z = (z_1, z')$ that make the metric Euclidean at x_P and $\nabla h(x_P) = s\partial_{z_1}$. The preparation bound and the common connection bound give $|D(\mathrm{d}h)| \le C|\mathrm{d}h|/S$. Integration along parent segments of length $O(\nu S)$ gives

$$\mathrm{d}h/s = \mathrm{d}z_1 + O(\nu), \qquad G_{ij} = I + O(\nu), \qquad |\nabla h|/s = 1 + O(\nu). \tag{3.6}$$

Rescale χ to the parent and set

$$v = h + s\ell a \chi \psi(z'/\ell). \tag{3.7}$$

The ratio between the pattern scale and cutoff scale is ν; hence $\mathrm{d}v/s = \mathrm{d}z_1 + a\chi f \sigma \cdot \mathrm{d}z' + O(\nu)$. This proves the gradient comparison.

We give the Hessian check also where the favorable coefficient vanishes. Put $D = \nabla v/\sqrt{1+|\nabla v|^2}$ and

$$\mathcal{F} = \{e : \mathrm{d}v(e) = 0,\ \sigma \cdot e' = 0\}.$$

This is a two-plane. For metric-unit $e \in \mathcal{F}$,

$$|e'|^2 \ge 1/2, \qquad |\sigma \cdot D'| \le C_0 a\chi f + C\nu.$$

After dividing the new Hessian test by s/ℓ, its principal contribution from (3.5) is

$$a\chi\{f'(\sigma \cdot D')^2 + u_0(|\Pi_{\sigma^\perp} D'|^2 + |e'|^2)\}.$$

Choose a so that $2C_0^2 a^2 \sup_v vF(v)|F'(v)| \le 1/4$. The displayed expression is then at least $a\chi u_0/4 - Ca\nu^2\chi|f'|$. Mixed cutoff terms on e, e vanish. On D, D they are bounded by

$$C\nu f |D_{\mathrm{sc}}\chi|(\chi f + \nu).$$

The remaining cutoff and connection errors are bounded by

$$C\nu^2|\psi||D^2_{\mathrm{sc}}\chi| + C\ell(\chi f + \nu).$$

Here D_{sc} denotes differentiation at the parent cutoff scale.

Let $X = \chi u_0$. On the compact set of cutoff and period positions and bounded affine frames, every continuous $b \ge 0$ that vanishes when $X = 0$ satisfies

$$\sup \frac{\nu^2 b}{X + \nu^2} \longrightarrow 0.$$

For $X \ge \tau$, bound the ratio by $\nu^2 \sup b/\tau$; for $X < \tau$, bound it by $\sup_{X<\tau} b$, then let $\tau \downarrow 0$. Apply this to $\chi|f'|$, $f|D_{\mathrm{sc}}\chi|$, $|\psi||D^2_{\mathrm{sc}}\chi|$; their vanishing follows from flatness and $u_0 = 0 \Rightarrow f = f' = \psi = 0$. The other mixed term is $O(\nu X)$, and $\chi f \le C\sqrt{X}$, $\ell \le \nu^2$ imply $\ell(\chi f + \nu)/(X + \nu^2) = O(\nu)$. Consequently, uniformly over admissible parents,

$$\frac{\ell}{s}\{\nabla^2(v-h)(D,D) + \nabla^2(v-h)(e,e)\} \ge c_a X - \varepsilon(\nu)(X + \nu^2), \qquad \varepsilon(\nu) \to 0. \tag{3.8}$$

The old Hessian is $O(\nu^2)$ in these normalized units. If $X/\nu^2 \to \infty$, the new positive term dominates it. If this ratio stays bounded, the old and new gradients differ by a relative $O(\nu)$. The map $T(p) = p/\sqrt{1+|p|^2}$ obeys

$$|T(q) - T(p)| \le 2\epsilon \quad \text{when } |q-p| \le \epsilon|p|, \quad \epsilon \le 1/2.$$

This follows by integrating its derivative along a segment whose distance from zero is at least $|p|/2$. A rotation $R = I + O(\nu)$ takes $\ker \mathrm{d}h$ to $\ker \mathrm{d}v$. Inside this three-dimensional space, RE and $\mathcal{F}$ have intersection dimension at least $2 + 2 - 3 = 1$. In that direction the old positive test changes by $o(\nu^2)$; the new contribution in (3.8) is bounded below by $-o(\nu^2)$. Thus a positive margin of size $c_{\mathrm{p}}\nu^2/8$ remains. A contrary sequence of configurations has a subsequence with bounded ratio or ratio tending to infinity; both cases exclude its failure. This proves the uniform one-direction margin on the whole parent. The intersection calculation is the use of base dimension four.

Where $\chi = 1$ and the period position lies in a core, u_0 has a fixed positive lower bound. Both directions in $\mathcal{F}$ then have test at least $c's/\ell$, while the Hessian norm is at most $(C' + O(C_{\mathrm{p}}\nu^2))s/\ell$. Fix sufficiently large C_{p} and small c_{p}, then sufficiently small ν; this region is prepared at scale ℓ. Retain only whole child cubes contained in it.

To compare mass, tile the affine coordinates by period cells of side ℓ, including a dummy z_1 period. Cells meeting the inner-parent boundary occupy relative volume $O(\nu)$. Discarding child cubes meeting core boundaries loses another $O(\nu)$ fraction, because the finite cores have finite boundary area in a period. On the retained region

$$|\nabla v|/s = \sqrt{1 + a^2 f^2} + O(\nu).$$

The fixed affine Jacobian cancels against parent volume. Thus

$$\frac{1}{s|P|}\int_{\mathcal{R}_P} |\nabla v| \,\mathrm{d}x \ge q_{\mathrm{in}} C_{\mathrm{core}} - C\nu, \qquad \frac{1}{s|P|}\int_P |\nabla h| \,\mathrm{d}x \le 1 + C\nu.$$

Taking both errors below η_0 gives a ratio exceeding $(1+3\eta_0)/(1+\eta_0) > 1+\zeta$, with $\zeta = \eta_0/(1+\eta_0) > 0$.

Finally, j derivatives of (3.7), with q on the cutoff, cost

$$s\ell(\nu S)^{-q}\ell^{-(j-q)} = s\ell^{1-j}\nu^q \le s\ell^{1-j}.$$

The slope and affine frame are frozen constants in this expression. This proves (3.4). The order of choices is the pattern and its small amplitude, the packing and cutoffs, the preparation constants, and finally ν. □

3.2 Iteration with every finite derivative order

Proof of Proposition 3.1. Start from one seed cube of side ν, prepared at scale 1, compactly inside U_0. Write h_0 for the seed and P_0 for this cube. Fix $s_{\min} > 0$ with $|\nabla_G h_0|_G \ge s_{\min}$ on P_0, common over $G \in \mathcal{G}$; this follows from the gradient separation on U_0 and the common metric bounds. Put $r_{\mathrm{c}} = \nu^2$. At generation i, apply Lemma 3.2 on each selected parent at scale $S_i = r_{\mathrm{c}}^{i-1}$. The selected children have side νr_{c}^i. Supports lie strictly inside parents, so every finite generation is smooth. The total selected gradient mass I_i satisfies

$$I_i \ge (1+\zeta) I_{i-1}, \qquad \int_V \sqrt{1 + |\nabla h_i|^2} \,\mathrm{d}x \ge I_i.$$

The initial I_0 has a common positive lower bound.

Gradients grow by at most β each generation and retain at least a factor α on the initial cube. Disjoint supports within a generation and (3.4) give

$$\|D^j(h_i - h_{i-1})\|_\infty \le C_j \beta^{i-1} r_{\mathrm{c}}^{i(1-j)}.$$

There is no factor counting cubes. The increments are added, so their higher derivative constants are not multiplied between generations. Summing yields one C_*, independent of j, with

$$\|h_m\|_{C^j} \le C_j e^{C_*(j+1)m}. \tag{3.9}$$

For example, take C_* larger than $\log\beta + |\log r_{\mathrm{c}}| + 1$. At a point last changed at generation $i \le m$, the retained Hessian margin is at least

$$c\, s_{\min} \alpha^{i-1} / r_{\mathrm{c}}^{i-1} \ge c\, s_{\min} \alpha^m.$$

Later modifications vanish there, with all jets vanishing at support boundaries. Outside the initial cube the seed is unchanged. Enlarge C_* to obtain $\max_e J_{h_m}(e) \ge c e^{-C_* m}$ on $t \ge 1/4$.

Choose once $\theta > 0$ with $C_*\theta \le \delta$, and put

$$m = \lfloor \theta \log k \rfloor, \qquad \gamma = \theta \log(1+\zeta) > 0, \qquad h_k = h_m + c_k.$$

The derivative and Hessian bounds follow from (3.9), and the mass bound follows from $(1+\zeta)^m \ge (1+\zeta)^{-1} k^\gamma$. All changes lie in the seed cube above $t = 1/2$. The constants determining C_*, θ, γ use the fixed pattern and common C^1 metric bounds only. A later smooth coordinate change contributes finite order-dependent chain-rule constants and lower derivatives of h_k, without changing these exponents. □

4 Local complex waves below the envelope

Fix a smooth positive input pair and an envelope from Proposition 3.1. A wave packet will have amplitude comparable to e^{kh} at its center, decay rapidly relative to that envelope away from the center, and have a residual as small as any prescribed inverse power of k. The finite construction must retain the same derivative and spanning exponents when its accuracy is increased.

4.1 Characteristic Hessians and polynomial equations

In G-normal coordinates at a center, let $p = \nabla h$, $Q = \sqrt{1+|p|^2}$ and $H = \nabla^2 h$. For a unit $v \perp p$, the bilinear square of $P = p + iQv$ is -1. Require the phase to satisfy the bilinear eikonal equation $G^{-1}(\mathrm{d}\phi, \mathrm{d}\phi) = -1$ to first order at the center. A phase with differential P must then have Hessian S satisfying $SP = 0$: differentiating the equation gives this condition because the first metric derivatives vanish at the center. The real Hessian gap $H - \operatorname{Re} S > 0$ will keep its exponential below e^{kh}.

Lemma 4.1. *Let e_1, v be orthonormal, $0 \le b < 1$, and H real symmetric, with*

$$|H| \le k^{4\delta}, \qquad b^2 H(e_1, e_1) + H(v, v) \ge k^{-2\delta}.$$

For all sufficiently large k, uniformly under these hypotheses, there are real symmetric U, D with

$$(U + iD)(be_1 + iv) = 0, \quad H - U \ge k^{-3\delta} I, \quad |U + iD| \le k^{20\delta}. \tag{4.1}$$

Proof. For $b > 0$ the first identity is $Dv = bUe_1,\ De_1 = -Uv/b$. The prescribed columns are compatible with symmetry exactly when $b^2U_{11} + U_{vv} = 0$. If $b \geq k^{-12\delta}$, put

$$c_0 = \frac{b^2H_{11} + H_{vv}}{1 + b^2}, \qquad U = H - c_0 I.$$

Relative to $e_1, v, \{e_1, v\}^\perp$, a symmetric completion is

$$D = \begin{pmatrix} -u_{12}/b & bu_{11} & -u_{2E}/b \\ bu_{11} & bu_{12} & bu_{1E} \\ -u_{E2}/b & bu_{E1} & 0 \end{pmatrix}.$$

Here the u's are the corresponding blocks of U. Then $H - U \geq \frac{1}{2}k^{-2\delta}I$ and $|D| \leq Ck^{16\delta}$.

For $0 \leq b < k^{-12\delta}$, set $K = k^{14\delta}$ and

$$U = -K(I - vv^{\mathsf{t}}) + b^2Kvv^{\mathsf{t}}, \qquad D = -bK(e_1v^{\mathsf{t}} + ve_1^{\mathsf{t}}).$$

The identity holds also at $b = 0$. In $H - U - k^{-3\delta}I$, the $v^\perp$ block is at least $KI/2$, while the scalar v block is at least

$$k^{-2\delta} - k^{-20\delta} - k^{-10\delta} - k^{-3\delta}.$$

Its Schur complement subtracts at most $C|H|^2/K \leq Ck^{-6\delta}$. All negative terms are smaller than $k^{-2\delta}$ for large k. The size bounds follow from the explicit matrices. □

Only the Hermitian size of a complex direction is needed to solve the subsequent polynomial equations. To see this, let $\mathcal{P}_m$ be the homogeneous complex polynomials of degree m, and suppose $c \leq |q| \leq C$. Choose $|q_i| \geq c/2$, write $q \cdot \partial = q_i\partial_i + D'$, and let I_i integrate in X_i with zero constant. The operator $\mathcal{K} = D'I_i/q_i$ lowers the degree in the other variables, so $\mathcal{K}^{m+1} = 0$ on $\mathcal{P}_m$. Thus

$$R_{q,m} = \frac{I_i}{q_i}\sum_{r=0}^{m}(-\mathcal{K})^r, \qquad (q \cdot \partial)R_{q,m} = I. \tag{4.2}$$

Its coefficient norm is bounded in terms of m, c, C. There is no division by the possibly small bilinear square $q \cdot q$.

4.2 Four packets with arbitrary finite accuracy

For packets centered at z, define their normalized first jets in that center's fixed normal coordinates by

$$\mathbf{j}_z w(x) = e^{-kh(x)}\left(w(x), \frac{\partial_\xi w(x)}{kQ_z}\right), \qquad Q_z = \sqrt{1 + |\nabla h(z)|^2}.$$

The real and imaginary columns of these jets form a real matrix. Its least spanning singular value means the square root of the least eigenvalue of the matrix times its transpose.

Proposition 4.2. *Given an integer $J \geq 0$ and $M > 0$, finite construction orders can be fixed so that, for every sufficiently large k, every center z with $t(z) \geq 1/4$ has four smooth complex packets. They have support radius at most $Ck^{-1/3-100\delta}$, and*

$$|D_x^j w| \leq C_j k^{2j} e^{kh} \qquad (j \geq 0), \tag{4.3}$$

$$|D_x^j (L + k^2)w| \leq Ck^{-M} e^{kh} \qquad (0 \leq j \leq J). \tag{4.4}$$

The family spans real first jets with singular value at least $ck^{-3/4}$ at z, and at least k^{-1} at distance at most k^{-10} from z. The support, derivative and spanning exponents do not depend on J, M. Constants and frequency thresholds may depend on the fixed pair and chosen orders.

Proof. The envelope bounds give $Q \le k^{3\delta}$ and $|H| \le k^{4\delta}$ for large k. If $p \neq 0$, take $e_1 = p/|p|$; choose a favorable test e_2 in (3.1). At $p = 0$, choose e_2 first and then any perpendicular e_1. Complete an orthonormal basis and put

$$\eta_{\mathrm{v}} = k^{-9\delta}, \quad v_2 = e_2, \quad v_3 = \frac{e_2 + \eta_{\mathrm{v}} e_3}{\sqrt{1+\eta_{\mathrm{v}}^2}}, \quad v_4 = \frac{e_2 + \eta_{\mathrm{v}} e_4}{\sqrt{1+\eta_{\mathrm{v}}^2}}.$$

Use the four pairs

$$(p', v) = (p, v_2), (p, v_3), (p, v_4), (p + Qk^{-3/4}e_1, v_2).$$

For each set $Q' = \sqrt{1+|p'|^2}$, $P = p' + iQ'v$, $b = |p'|/Q'$. Then $P \cdot P = -1$ and $1 \le Q'/Q \le 1 + k^{-3/4}$. The change in the favorable Hessian test is at most $Ck^{-5\delta} + Ck^{-3/4+4\delta}$. For the second term, differentiate $s^2/(1+s^2)$ as $s = |p|$ increases by $Qk^{-3/4}$. These errors are smaller than the original $ck^{-\delta}$ margin. Lemma 4.1 therefore gives U, D with $(U + iD)P = 0$ for each pair.

Each packet uses one frozen normal chart $\Psi_{z,E}(\xi) = \exp_z(\sum_i \xi_i E_i)$. For a fixed smooth input metric, all fixed-order derivatives of these charts and their inverses are uniformly bounded over centers and unit frames on sufficiently small common balls. This follows by solving the smooth geodesic equation on a common bounded set and differentiating it in initial conditions, then using compactness of the frame bundle. No spatial derivative of the frame-selection rule is taken.

Introduce the scaled variables

$$r_0 = k^{-100\delta}, \quad X = \xi/r_0, \quad \kappa = kr_0Q, \quad h_{\mathrm{sc}}(X) = \frac{h(\Psi_{z,E}(r_0X)) - h(z)}{r_0Q}. \tag{4.5}$$

For $q \ge 2$, the chain rule gives

$$|D_X^q h_{\mathrm{sc}}| \le C_q Q^{-1} k^{\delta(101-99q)}.$$

These are bounded for every fixed order. If $L = g^{ij}(\xi)\partial_{\xi_i\xi_j} + \beta^i(\xi)\partial_{\xi_i}$, multiplying $L + k^2$ by r_0^2 gives

$$\mathcal{L} + \kappa^2/Q^2, \qquad \mathcal{L} = g^{ij}(r_0X)\partial_{X_iX_j} + r_0\beta^i(r_0X)\partial_{X_i}.$$

Its fixed-order coefficient derivatives are bounded, with $g^{ij}(0) = \delta_{ij}$ and $\partial g^{ij}(0) = 0$.

For an integer $T \ge 3$, construct a polynomial phase with

$$\Phi(0) = 0, \quad \partial\Phi(0) = P/Q, \quad \partial^2\Phi(0) = r_0(U + iD)/Q$$

so that $E_\Phi = g^{ij}(r_0X)\partial_i\Phi\,\partial_j\Phi + Q^{-2}$ vanishes to order T at zero. The constant and linear equations are already satisfied. At homogeneous degree $m \ge 2$, the new term Φ_{m+1} enters as $2(P/Q)\cdot\partial\Phi_{m+1}$; everything else is known. Its direction has Hermitian norm between 1 and 3, so (4.2) solves every equation. For fixed T all coefficients are uniformly bounded.

Put

$$\mathcal{T}a = 2g^{ij}(r_0X)(\partial_i\Phi)(\partial_j a) + (\mathcal{L}\Phi)a.$$

For a finite integer $s \ge 0$, solve by homogeneous degrees for polynomials $a_0, \dots, a_s$, with $a_0(0) = 1$, $a_j(0) = 0$ for $j \ge 1$, such that

$$\mathcal{T}a_0, \qquad \mathcal{T}a_j + \mathcal{L}a_{j-1} \quad (1 \le j \le s)$$

vanish to order T. Again the new degree-$(m+1)$ term enters via $2(P/Q)\cdot\partial$. Complete all of a_{j-1} before starting a_j, so the second derivatives in its right-hand side are already determined. Every coefficient is bounded for fixed T, s.

Choose a real smooth cutoff ω, equal to one on the unit ball and supported in the radius-two ball, and define

$$a = \sum_{j=0}^{s} \kappa^{-j} a_j, \qquad w(\Psi_{z,E}(r_0 X)) = e^{kh(z)} e^{\kappa\Phi(X)} a(X)\omega(k^{1/3}X). \tag{4.6}$$

It extends smoothly by zero and has the stated support radius. On its support $a = 1 + O(k^{-1/3})$, hence $|a| \geq 1/2$.

The real phase lies below the envelope to quadratic order:

$$\operatorname{Re}\Phi - h_{\mathrm{sc}} \leq k^{-3/4}|X| - ck^{-106\delta}|X|^2 + C|X|^3. \tag{4.7}$$

The linear term is zero for the first three packets. On $|X| \leq 2k^{-1/3}$, the cubic term is absorbed into half the quadratic deficit. The remaining maximum is $O(k^{-3/2+106\delta})$; since $k^{1-100\delta} \leq \kappa \leq k$, the exponential in (4.6) is at most $2e^{kh}$. On the annulus $\frac{1}{4}k^{-1/3} \leq |X| \leq 2k^{-1/3}$ it is at most

$$e^{-2k^{1/4}} e^{kh}. \tag{4.8}$$

Indeed, the negative exponent has size at least $ck^{1/3-206\delta}$, and the positive linear exponent is at most $Ck^{-1/12}$.

Direct differentiation before the cutoff gives

$$\begin{aligned} e^{-\kappa\Phi}(\mathcal{L} + \kappa^2/Q^2)(e^{\kappa\Phi}a) = \kappa^2 E_\Phi a + \kappa \mathcal{T} a_0 \\ + \sum_{j=1}^{s} \kappa^{1-j}(\mathcal{T} a_j + \mathcal{L} a_{j-1}) + \kappa^{-s}\mathcal{L} a_s. \end{aligned}$$

Taylor remainders and their $q \leq J \leq T$ derivatives cost $|X|^{T-q}$. In ordinary coordinates a derivative of such a remainder costs at most $k^{1/3+100\delta}$; derivatives of the exponential, amplitude and cutoff cost at most Ck^2 each. Undoing the operator scaling gives the bound

$$Ck^{200\delta+2J}\left(k^{2-T/3} + k^{1-T/3} + k^{-(1-100\delta)s}\right) e^{kh}. \tag{4.9}$$

All cutoff commutators have polynomial factors times (4.8). Choose, before k,

$$T > 3(M + 200\delta + 2J + 3), \qquad s > \frac{M + 200\delta + 2J + 1}{1 - 100\delta}.$$

Increasing the frequency threshold then proves (4.4). The same calculation without the operator factor proves (4.3), with an exponent independent of T, s.

At the center, $\mathbf{j}_z w = (1, P/Q) + O(\kappa^{-1})$. The real leading columns contain $(1, (|p|/Q)e_1)$ and $(1, (|p|/Q + k^{-3/4})e_1)$; they span the value–$e_1$ plane with singular value at least $ck^{-3/4}$. The imaginary columns span $e_1^{\perp}$ with singular value at least $ck^{-9\delta}$. The amplitude error is $O(k^{-1+100\delta}) = o(k^{-3/4})$. Through two derivatives (4.3) also gives $|D_x \mathbf{j}_z w| \leq Ck^3$. Moving by k^{-10} changes the matrix by only $O(k^{-7})$. Fixed coordinate changes have bounded inverses, so the nearby lower bound is k^{-1} for large k. Every step includes $p = 0$. □

4.3 Estimates on wavelength balls

The next estimates control all packets that meet a small ball, including packets centered outside the region where signs will be measured. This is needed because a value-normalized Gaussian sum must have bounded local derivative moments.

Lemma 4.3. *Fix the packet orders. Let $\mathcal{W}$ contain all four packets at each of finitely many centers, with $|\mathcal{W}| \le Ck^{K_1}$ for fixed K_1. Suppose every $x \in V$ has a center within C_0k^{-10}. Define, in the fixed chart Ω,*

$$R_x = (kQ_x)^{-1}, \quad S_x^2 = \sum_{w\in\mathcal{W}} |w(x)|^2, \quad w_x(Y) = w(x + R_xY), \quad |Y| \le 2.$$

For packets whose support meets this ball, let X_z denote their own centered scaled coordinates. Call them non-tail when $|X_z(x)| < \frac{1}{2}k^{-1/3}$, and tail otherwise. For every fixed $j \ge 0$,

$$S_x \ge ce^{kh(x)}, \tag{4.10}$$

$$\sup_{|Y|\le 2} |D_Y^j w_x(Y)| \le C_j|w(x)| \qquad \textit{for non-tails}, \tag{4.11}$$

$$\sup_{|Y|\le 2} |D_Y^j w_x(Y)| \le C_j e^{-k^{1/4}} e^{kh(x)} \qquad \textit{for tails}. \tag{4.12}$$

Every non-tail has nonzero center value and cutoff identically one on the ball, and

$$\left|\operatorname{Im} \frac{\mathrm{d}_Y w_x}{w_x}(0)\right| \ge c. \tag{4.13}$$

Consequently,

$$S_x^{-2} \sum_w \sup_{|Y|\le 2} |D_Y^j w_x|^2 \le C_j, \qquad S_x^{-2} \sum_{\textit{non-tail } w} |w(x)|^2 = 1 - o(1). \tag{4.14}$$

Constants and thresholds may depend on the fixed pair, orders and K_1, C_0, but not on k, x.

Proof. The envelope gives $|DQ| \le Ck^{4\delta}$. Any intersecting packet center has

$$|z - x| \le Ck^{-1/3-100\delta} + C/k, \qquad |Q_z - Q_x| \le Ck^{-1/3-96\delta} + Ck^{-1+4\delta} = o(1).$$

All such centers and balls lie in a common compact subchart for large k. Since $Q \ge 1$, the ratios Q_z/Q_x and the corresponding ratios throughout the ball are bounded above and below. From $|\mathrm{d}h| \le CQ$,

$$k|h(x + R_xY) - h(x)| \le C \quad (|Y| \le 2).$$

The nearest center has $|X_z(x)| = O(k^{-10+100\delta})$. Its center value and nearby jet estimates yield $|w(x)| \ge ce^{kh(x)}$ for one of its packets, proving (4.10).

With $\kappa_z = kr_0Q_z$, fixed normal-chart bounds give

$$|D_Y^q X_z| \le C_q r_0^{-1} R_x^q, \qquad \kappa_z |D_Y^q X_z| \le C_q (Q_z/Q_x)(kQ_x)^{1-q} \le C_q \quad (q \ge 1).$$

For $q = 1$, $\kappa_z D_Y X_z = (Q_z/Q_x) D_x \xi_z$ has a uniformly bounded inverse. The displacement of X_z on the ball is $O(k^{-1+100\delta}/Q_x) = o(k^{-1/3})$.

For a tail, any point where the cutoff or its derivatives contribute satisfies $\frac{1}{4}k^{-1/3} \le |X_z| \le 2k^{-1/3}$. Use (4.8). Each fixed positive Y-derivative of $\kappa_z\Phi(X_z)$ is bounded by the preceding

estimates; products with m derivatives of X_z leave $\kappa_z^{1-m} \le 1$. Amplitude derivatives are bounded, and $k^{1/3}/\kappa_z \le k^{-2/3+100\delta} \le 1$ controls cutoff derivatives. The exponential estimate therefore gives (4.12), even when $w(x) = 0$.

For a non-tail the cutoff stays one on the ball and the amplitude is bounded away from zero. The same derivative bounds control the logarithmic derivative and all higher derivatives relative to $|w_x|$. Integrating its real logarithmic derivative along straight segments gives $|w_x(Y)| \le C|w(x)|$, proving (4.11). Moreover,

$$\operatorname{Im} \partial_X \Phi(X_z(x)) = (Q'/Q_z)v + O(k^{-1/3}).$$

Its leading norm is at least one, so the inverse bound on $\kappa_z D_Y X_z$ gives a positive imaginary Y-gradient. The amplitude's logarithmic derivative is only $O(k^{-1+100\delta}/Q_x) = o(1)$, which proves (4.13). Squaring the tail bounds and using (4.10) bounds their total normalized contribution by $C_j k^{K_1} e^{-2k^{1/4}} = o(1)$. The non-tail contributions are controlled by their center values. The definition of S_x then proves (4.14). □

5 An exact simple eigenfunction after a small perturbation

Fix the input pair, l_-, l_+, and put $w_{\mathrm{c}} = l_+ - l_- > 0$, $s_i = l_- + iw_{\mathrm{c}}/5$ for $1 \le i \le 4$. For the allowed $k = \sqrt{N(N+3)}$, let

$$b_N(t) = \frac{N}{2k} \log(1-t).$$

Choose the additive constant in the envelope so that $h(s_2) = b_N(s_2)$. It is bounded uniformly in N. For the fixed small a_*, and all sufficiently large N,

$$\partial_t(h - b_N) = -a_* + \frac{N}{2k(1-t)} \ge c_0 > 0 \quad (0 \le t \le 1/2).$$

It follows that

$$h \le b_N - cw_{\mathrm{c}} \ (t \le s_1), \qquad h \ge b_N + cw_{\mathrm{c}} \ (s_3 \le t \le s_4). \tag{5.1}$$

Take $0 \le \beta \le 1$, equal to one for $t \le s_3$ and zero for $t \ge s_4$. The smooth positive weight

$$W = e^{kh} + \beta(t) e^{kb_N(t)} \tag{5.2}$$

uses a zero extension of the second summand. The pair is round on the support of β. Thus $(L + k^2)(\beta u_*)$ is only a cutoff commutator. Its support lies in the exponential gap of (5.1). For every fixed $q \ge 0$ and $M > 0$,

$$|D^q(\beta u_*)| \le C_q k^q W, \quad |D^q(L + k^2)(\beta u_*)| \le C_{q,M} k^{-M} W, \quad |\mathrm{d}W|/W \le Ck^2. \tag{5.3}$$

Derivatives of the fixed collar cutoff can depend on w_{c}; the positive gap absorbs them when k is sufficiently large.

5.1 A real sample with a nonvanishing first jet

Choose a net in $\{t \ge s_1\}$ of mesh at most k^{-10}. There are at most Ck^{40} centers. At each take the four packets of Proposition 4.2, and denote the collection by $\mathcal{W}_k$. Use independent standard real Gaussian coefficients $g_v^{(1)}, g_v^{(2)}$ and set

$$u = \beta u_* + \sum_{v \in \mathcal{W}_k} (g_v^{(1)} \operatorname{Re} v + g_v^{(2)} \operatorname{Im} v). \tag{5.4}$$

Let $\mathcal{B}_k$ be the event that all coefficients have absolute value at most k. The elementary Gaussian tail bound gives $\mathbb{P}(\mathcal{B}_k^c) \le Ck^{40}e^{-k^2/2} \to 0$.

Given finite $J \ge 2$ and $M > 0$, choose the packet residual orders through J to exceed $M + 42$. On $\mathcal{B}_k$, their number, (4.3) and (5.3) give

$$|D^q u| \le C_q k^{2q+41} W, \qquad |D^q \mathfrak{r}| \le C_q k^{-M} W \quad (0 \le q \le J), \qquad \mathfrak{r} = \operatorname{div}_\mu(A\,\mathrm{d}u) + k^2 \rho u. \tag{5.5}$$

All orders are chosen before k. Every packet vanishes for $t < s_1 - Ck^{-1/3-100\delta}$, so in that region $u = u_*$ and $\mathfrak{r} = 0$.

Lemma 5.1. *For every fixed choice of packet orders,*

$$\mathbb{P}\left(\mathcal{B}_k \cap \left\{\inf_B \frac{|u| + |\mathrm{d}u|_{g_0}}{W} \ge k^{-500}\right\}\right) \longrightarrow 1.$$

Proof. Where $W/e^{kh} > k^{50}$, the cutoff is one for large k, by (5.1). The cosine and angular derivative of the uncut beam give $|u_*| + |\mathrm{d}u_*|_{g_0} \ge ce^{kb_N}$. On $\mathcal{B}_k$ the packet first jets are at most $Ck^{43}e^{kh}$, so the beam dominates there and gives a lower bound cW. This region includes $t < s_1$.

In the remaining region there is a nearby center. The real jet spanning from Proposition 4.2, converted to ordinary derivatives, gives a covariance matrix for the five-dimensional value–gradient vector whose least eigenvalue has square root at least

$$ck^{-1}e^{kh} \ge ck^{-51}W \ge k^{-52}W.$$

One packet family suffices; others add nonnegative covariance. The deterministic beam shifts only the mean. Hence the density of the normalized vector $\mathcal{J} = W^{-1}(u, \partial_1 u, \dots, \partial_4 u)$ is at most Ck^{260}. For fixed $C_0, D > 0$,

$$\mathbb{P}(|\mathcal{J}(x)| \le C_0 k^{-D}) \le Ck^{260-5D}. \tag{5.6}$$

This is an unconditional Gaussian estimate.

On $\mathcal{B}_k$, two derivatives in (5.5) and $|\mathrm{d}W|/W \le Ck^2$ give $|D\mathcal{J}| \le Ck^{45} \le k^{46}$. In a finite atlas take grids of mesh k^{-D-48}, with at most Ck^{4D+192} points. Apply (5.6) at the Gaussian-region grid points and the beam bound elsewhere. The union bound is Ck^{452-D}. With $D = 500$ it tends to zero, while the interpolation error is $O(k^{-D-2})$. Choose C_0 to cover the fixed coordinate norm comparisons. This yields the stated global first-jet bound, after also subtracting $\mathbb{P}(\mathcal{B}_k^c)$. The use of $\mathcal{B}_k$ is confined to derivative control and interpolation, not to the Gaussian density calculation. □

5.2 Many strict signs at the local wavelength

Lemma 5.2. *For fixed packet orders there are $p_*, c_* > 0$ such that, for all sufficiently large allowed k, u has a sign certificate of size at least $c_* k^{1+\gamma}$ with probability at least p_*. Potential disks can be fixed deterministically before sampling.*

Proof. In V, use radii $R_x = (kQ_x)^{-1}$. Since $|DQ| \le Ck^{4\delta}$, intersecting balls have comparable Q's and radii for large k. At a fixed k these radii have a positive minimum. Choose a finite maximal disjoint family $\{B(x, R_x) : x \in \mathcal{X}_k\}$. Fixed enlargements cover V, and Q is comparable to Q_x in each enlargement. Therefore

$$\int_V kQ\,\mathrm{d}x \le C \sum_{x \in \mathcal{X}_k} kQ_x R_x^4 = C \sum_x R_x^3, \qquad \sum_x R_x^3 \ge ck^{1+\gamma}. \tag{5.7}$$

The latter inequality uses the integral in (3.1).

The beam vanishes on radius-two enlargements of these balls. On one define

$$S_x^2 = \sum_v |v(x)|^2, \qquad U_x(Y) = u(x + R_x Y)/S_x, \quad |Y| \le 2.$$

Lemma 4.3 gives all fixed-order squared derivative bounds after this normalization, and the non-tail value fraction is $1 - o(1)$. The value $U_x(0)$ is a standard real Gaussian.

Let z be the coefficient vector with pair $(\operatorname{Re} v(x), \operatorname{Im} v(x))$ for each packet, and T the matrix of their four real Y-derivatives at zero. Then $|z| = S_x$, and the gradient covariance conditional on the value is

$$S_x^{-2} T \left(I - \frac{z z^{\mathsf{t}}}{S_x^2} \right) T^{\mathsf{t}}. \tag{5.8}$$

The formula follows by an orthogonal decomposition of the standard Gaussian coefficient vector along z. For each non-tail, the unit coefficient direction

$$c_v = (-\operatorname{Im} v(x), \operatorname{Re} v(x))/|v(x)|$$

is perpendicular to z; these directions are mutually orthonormal, and

$$T c_v = |v(x)| \operatorname{Im} \frac{\mathrm{d}_Y v}{v}(0).$$

Equations (4.13) and (4.14) give a fixed positive trace lower bound in (5.8). There is a deterministic unit direction e_x such that $Z_x = \partial_{e_x} U_x(0)$, conditional on $U_x(0)$, has variance between fixed positive constants. Unconditional variance and the regression coefficient are bounded above by the derivative square sums.

Those same estimates through order five, integrated on $B(0,2) \subset \mathbb{R}^4$, give

$$\mathbb{E}\|U_x\|^2_{C^2(\overline{B(0,1)})} \le C\, \mathbb{E}\|U_x\|^2_{H^5(B(0,2))} \le C. \tag{5.9}$$

The Sobolev inequality here is the $k = 5, m = 2, p = 2, n = 4$ case of [4, Theorem 3.49, p. 76]. In particular, it controls the Euclidean gradient and Hessian operator norms.

Take $a' = 1/4$, $0 < c' < a'/4$. For each sufficiently large fixed H, the conditional Gaussian density gives

$$\mathbb{P}(1 \le Z_x \le 2,\ |U_x(0)| < c'/H) \ge c_1/H.$$

Its conditional mean is bounded on the indicated value interval, and its conditional variance stays above and below fixed positive numbers. Markov's inequality applied to (5.9) costs at most C/H^2 for the failure of $\|U_x\|_{C^2} \le H$. Fix H large enough so the intersection has probability $\pi_* = c_1/(2H) > 0$.

Taylor's formula at $\pm a' e_x/H$ now gives opposite values of absolute size at least $a'/(2H)$. The gradient bound keeps those signs on the parallel transverse disks of radius $\sigma_* = a'/(4H^2)$. Their joining cylinder lies inside the unit ball. The directions, disks and radii are deterministic; only success depends on the Gaussian sample.

Let F_k be the fraction of successful balls weighted by R_x^3. Then $0 \le F_k \le 1$, $\mathbb{E}F_k \ge \pi_*$. Writing $q = \mathbb{P}(F_k \ge \pi_*/2)$ gives $\pi_* \le (\pi_*/2)(1 - q) + q$, hence

$$\mathbb{P}(F_k \ge \pi_*/2) \ge \pi_*/(2 - \pi_*) > 0.$$

No independence between balls is needed. Each disk has three-area proportional to $\sigma_*^3 R_x^3$. Summing and using (5.7) proves the lemma. □

After all finite orders are fixed, Lemma 5.2 has one positive probability p_*, while failure in Lemma 5.1 tends to zero. For every sufficiently large allowed k, probability subtraction therefore gives a real sample satisfying (5.5), the global first-jet lower bound, and a certificate of size $c_* k^{1+\gamma}$.

5.3 Canceling the residual exactly

Lemma 5.3. *Suppose the first-jet bound in Lemma 5.1 holds. For an integer $R \geq 0$, assume the derivative bounds* (5.5) *for u through $R+2$ and for $\mathfrak{r}$ through $R+1$. There are smooth real corrections to the pair, zero on every open set where $\mathfrak{r} = 0$, such that*

$$\operatorname{div}_\mu((A+\delta A)\mathrm{d}u) + k^2(\rho+\delta\rho)u = 0, \qquad \|(\delta A, \delta\rho)\|_{C^R} \leq C_R k^{-M+2129+1088R}.$$

Thus $M \geq 2131 + 1088R$ gives corrections tending to zero in C^R.

Proof. Put $I = g_0^{-1}$, and define

$$\mathcal{D} = u^2 + I(\mathrm{d}u, \mathrm{d}u), \quad F = -\mathfrak{r}/\mathcal{D}, \quad \delta A = FuI, \quad \delta\rho = k^{-2}\big(Fu - \operatorname{div}_\mu(FI\mathrm{d}u)\big). \tag{5.10}$$

The first-jet bound implies $\mathcal{D} \geq \frac{1}{2}k^{-1000}W^2 > 0$. Consequently all four expressions are globally smooth. The product rule gives

$$\operatorname{div}_\mu(FuI\mathrm{d}u) + \big(Fu - \operatorname{div}_\mu(FI\mathrm{d}u)\big)u = F\big(u^2 + I(\mathrm{d}u, \mathrm{d}u)\big) = -\mathfrak{r}.$$

This is the required exact identity.

For $0 \leq q \leq R+1$, the bounds on u give

$$|D^q\mathcal{D}| \leq C_q k^{2q+86} W^2.$$

A derivative of order $q \geq 1$ of $\mathcal{D}^{-1}$ is a sum of terms $\mathcal{D}^{-m-1}\prod_{\nu=1}^m D^{q_\nu}\mathcal{D}$, where $q_\nu \geq 1$, $\sum q_\nu = q$, and $m \leq q$. The lower bound for $\mathcal{D}$ therefore yields, also at order zero,

$$|D^q\mathcal{D}^{-1}| \leq C_q k^{1000+1088q} W^{-2}, \qquad |D^q F| \leq C_q k^{-M+1000+1088q} W^{-1}.$$

These are estimates on already differentiated factors; they do not differentiate W^{-1}. Multiplication by derivatives of u cancels the weights and gives

$$|D^q(Fu)| \leq C_q k^{-M+1041+1088q}, \qquad |D^q(FI\mathrm{d}u)| \leq C_q k^{-M+1043+1088q}.$$

The divergence in (5.10) costs one derivative, followed by k^{-2}. Its exponent is $1043 + 1088(R+1) - 2 = 2129 + 1088R$, as stated. If $\mathfrak{r}$ vanishes on an open set, then F and its derivatives vanish there, so both corrections vanish there. For sufficiently large k they preserve positivity, since the input pair is positive on a compact manifold. □

5.4 Making the eigenvalue simple while retaining its eigenfunction

Lemma 5.4. *Suppose u is an exact real eigenfunction at $k^2 = N(N+3)$, with $N > 1$. Suppose the pair is round and $u = u_*$ on an open region containing a point where $0 < t < 1$ and $\mathrm{d}u, \mathrm{d}t$ are independent. A smooth tensor perturbation supported there can make k^2 simple while retaining u and ρ. Its size can be arbitrarily small in any prescribed finite C^R norm.*

Proof. Take a small product coordinate patch O with coordinates (u, t, z_3, z_4). Choose a nonnegative bump χ supported inside O, positive on a smaller product patch O_0. On O put

$$E = \chi \left(I - \frac{(I\mathrm{d}u) \otimes (I\mathrm{d}u)}{I(\mathrm{d}u, \mathrm{d}u)} \right), \qquad I = g_0^{-1},$$

and extend it by zero. The tensor is smooth, nonnegative and $E\mathrm{d}u = 0$. Let $\mathcal{E}$ be the old real eigenspace at k^2. We first show that the kernel on $\mathcal{E}$ of $\int_B E(\mathrm{d}f, \mathrm{d}f)\mathrm{d}\mu$ is exactly $\mathbb{R}u$.

If the integral is zero, positivity on O_0 gives $\mathrm{d}f \in \mathbb{R}\,\mathrm{d}u$ there. On its product coordinates this means $f = b(u)$, with b smooth. The round equations imply

$$b''(u)|\mathrm{d}u|^2_{g_0} = k^2(ub'(u) - b(u)). \tag{5.11}$$

The ambient gradient of $\mathrm{Re}(x_1 + ix_2)^N$ has squared norm $N^2(1-t)^{N-1}$ and radial derivative Nu. Projection to the tangent space of the sphere gives

$$|\mathrm{d}u|^2_{g_0} = N^2\big((1-t)^{N-1} - u^2\big).$$

Differentiate (5.11) in the coordinate t, holding u fixed. Since $N > 1$ and $t < 1$, this gives $b'' = 0$; substitution eliminates the constant term of b. Thus $f = cu$ on O_0. The difference satisfies a smooth uniformly elliptic second-order equation. Aronszajn's theorem [3, pp. 235–236] applies: the smooth principal coefficients satisfy its C^2 and Lipschitz-second-derivative requirements, and the lower-order coefficients are bounded locally. Open-set vanishing gives infinite-order vanishing at an interior point. Unique continuation, applied in overlapping charts on connected B, therefore gives $f = cu$ globally.

Now replace A by $A + \tau E$, with $\tau > 0$. The same u still solves the exact equation at k^2. If that eigenvalue were not simple for arbitrarily small positive τ, there would be $\tau_m \downarrow 0$ and corresponding real eigenfunctions f_m, of unit weighted L^2 norm and orthogonal to u. Their energy identity

$$\int_B (A + \tau_m E)(\mathrm{d}f_m, \mathrm{d}f_m)\mathrm{d}\mu = k^2$$

bounds them in H^1. Weak H^1 compactness and compact $H^1 \hookrightarrow L^2$ give a subsequence converging to a norm-one f_0 orthogonal to u. The weak equation passes to the limit; local elliptic regularity makes f_0 smooth, so $f_0 \in \mathcal{E}$. For any fixed $\phi \in \mathcal{E}$, testing the old equation against f_m and the perturbed one against ϕ, then subtracting and dividing by τ_m, gives

$$\int_B E(\mathrm{d}f_m, \mathrm{d}\phi)\mathrm{d}\mu = 0.$$

Weak convergence preserves this identity. Taking $\phi = f_0$ places f_0 in the kernel already shown to be $\mathbb{R}u$, which contradicts its norm and orthogonality. Thus every sufficiently small positive τ makes k^2 simple. The tensor E is fixed before this scalar is chosen, so τE can meet any finite derivative tolerance. The compactness and regularity used here are justified explicitly in Lemma 6.1 below. □

5.5 Completion of the quantitative step

Proof of Proposition 2.2. Fix the pair, collar, r, η. Choose $R \geq \max(r, 1)$ and a smaller positive tolerance η' such that movement less than η' in C^R implies the desired C^r bound and keeps the pair positive and inside $\mathcal{U}$. This is possible because the fixed input is in the open neighborhood.

Set $J = R + 2$, take $M \geq 2131 + 1088R$, and then choose the finite packet orders for (5.5). These choices precede the frequency.

For every sufficiently large allowed k, the simultaneous sample above has a certificate of size $c_* k^{1+\gamma}$ and the hypotheses of Lemma 5.3. Choose k sufficiently large that its correction is smaller than $\eta'/2$ in C^R. The protected round region remains nonempty for every fixed collar width: it suffices to require

$$Ck^{-1/3-100\delta} < w_{\mathrm{c}}/100, \qquad cw_{\mathrm{c}}k > (M' + P)\log k + \log C_q(w_{\mathrm{c}}). \tag{5.12}$$

The second inequality makes a cutoff error $C_q(w_{\mathrm{c}})k^P e^{-cw_{\mathrm{c}}k}$ smaller than $k^{-M'}$. There are only finitely many derivative orders and accuracy targets at this stage, so all these inequalities hold for every sufficiently large k. Below $s_1 - w_{\mathrm{c}}/100$, $u = u_*$ and the residual and corrections vanish.

In $l_- + w_{\mathrm{c}}/20 < t < l_- + w_{\mathrm{c}}/10$, choose a point where $\sin(N\vartheta) \neq 0$. Here $\mathrm{d}t \neq 0$ and the angular derivative of u_* is nonzero, so $\mathrm{d}u, \mathrm{d}t$ are independent. Apply Lemma 5.4. After k and the sample have been fixed, choose its scalar τ below the simplicity threshold with $\|\tau E\|_{C^R} < \eta'/2$. All derivatives of E are finite at this stage; their possible growth with k causes no obstruction to this later choice. The perturbation is supported in $t > l_-$. It preserves the function, eigenvalue and signs exactly.

The final pair is round for $t < l_-$, meets the norm tolerance, and has a simple eigenvalue and the required certificate. The number $c_* > 0$ is fixed before the universal choice of large frequency. The exponent γ, chart Ω and neighborhood $\mathcal{U}$ were fixed earlier, independent of all the finite orders, tolerances and collar widths. □

6 Persistence and one fixed smooth metric

6.1 A finite norm suffices to preserve earlier signs

Lemma 6.1. *Let a smooth positive pair $\mathcal{P}$ have a simple positive eigenvalue λ and a real eigenfunction with a sign certificate $\mathcal{C}$. For any open interval $I \subset (0, \infty)$ containing λ, there are an integer $s \geq 1$ and $\varepsilon > 0$ such that every smooth positive pair $\mathcal{P}'$ with $\|\mathcal{P}' - \mathcal{P}\|_{C^s} \leq \varepsilon$ has a simple eigenvalue in I and a real eigenfunction with the same certificate disks.*

Proof. We specify the compactness and regularity facts used both here and in Lemma 5.4. On $H^1(B)$, the shifted form

$$a(f, v) = \int_B A(\mathrm{d}f, \mathrm{d}v)\mathrm{d}\mu + \int_B \rho f v \mathrm{d}\mu$$

is continuous, symmetric and coercive. The inclusion $H^1(B) \hookrightarrow L^2(\rho\mathrm{d}\mu)$ is continuous, compact and dense. These properties follow in a finite atlas from the local Sobolev compactness theorem [4, Theorem 3.49, p. 76]; positive smooth densities give equivalent L^2 norms. Laugesen's discrete spectral theorem [5, Theorem 4.1, pp. 28–32], with form space H^1 and Hilbert space $L^2(\rho\mathrm{d}\mu)$, gives an orthonormal eigenbasis for the shifted problem with eigenvalues tending to infinity. Subtracting one gives the nonnegative discrete spectrum of $-L$, with finite multiplicities. All eigenfunctions can be chosen real.

For completeness, weak eigenfunctions are smooth. In a coordinate chart write $\mathrm{d}\mu = m(x)\mathrm{d}x$. The divergence coefficients are mA^{ab}, and the right-hand side is a smooth density times the eigenfunction. Local uniform ellipticity and [4, Theorems 4.27–4.28, pp. 112–114] first give H^2_{loc} and then successively every finite Sobolev order. Their hypotheses on coefficient derivatives and the Sobolev regularity of the right-hand side are met at each step. Sobolev embedding and a finite

cover give global smoothness. The same argument applies to a weak limit solving a fixed smooth eigenfunction equation.

The eigenbasis also gives the min–max characterization

$$\lambda_n = \inf_{\dim F = n+1} \sup_{0 \neq f \in F} \frac{\int_B A(\mathrm{d}f, \mathrm{d}f)\mathrm{d}\mu}{\int_B \rho f^2 \mathrm{d}\mu}, \qquad n = 0, 1, \ldots. \tag{6.1}$$

Indeed, the first $n+1$ eigenvectors give the upper bound. Every $(n+1)$-dimensional F contains a nonzero vector orthogonal to the first n, whose eigenbasis expansion gives the lower bound. The characterization is on the common form domain H^1, including when the density varies. Uniform multiplicative comparisons of nearby positive tensors and densities compare all the quotients in (6.1). Thus C^0 convergence of pairs implies convergence of every fixed ordered eigenvalue. A smaller interval with closure in I can separate the simple λ from its neighboring ordered eigenvalues. For all sufficiently close pairs it contains exactly the corresponding eigenvalue, which is still simple.

We need convergence of its normalized eigenfunction in C^0. The same interior estimates, localization by a finite smooth partition of unity and absorption of lower-order terms give, for the fixed smooth operator L,

$$\|f\|_{H^{a+2}} \leq C_a(\|Lf\|_{H^a} + \|f\|_{L^2}), \qquad a \geq 0. \tag{6.2}$$

One way to see the absorption is to apply the interior estimates on a slightly larger chart to cutoff multiples of f. The commutator with the cutoff is first order. The inequalities $\|f\|_{H^{a+1}} \leq \epsilon\|f\|_{H^{a+2}} + C_\epsilon\|f\|_{L^2}$ follow from the elementary Fourier weight inequality in each chart. A finite sum and sufficiently small ϵ absorb those commutators and give (6.2).

A sufficiently small C^4 change of the pair makes $L' - L$ small as a map $H^{a+2} \to H^a$ for $a = 0, 2$. In coordinates its principal coefficients have no coefficient derivatives, and its first-order coefficients have at most one; the stated finite norm bounds every product-rule coefficient needed at these two orders. Absorb $\|(L' - L)f\|_{H^a}$ into (6.2). For eigenvalues in the chosen bounded interval, the estimates at $a = 0$ and then $a = 2$ give uniform H^4 bounds for weighted-L^2-normalized eigenfunctions.

If $\mathcal{P}_m \to \mathcal{P}$ in this finite norm, choose real eigenfunctions f_m with $\int_B \rho_m f_m^2 \mathrm{d}\mu = 1$. A subsequence converges strongly in H^3, and hence in C^0; the dimension-four embedding $H^3 \hookrightarrow C^0$ again follows from Hunter's Theorem 3.49 in a finite atlas. The weak equation passes to the limit, whose norm is one because $\rho_m \to \rho$ uniformly. Simplicity identifies the limit with the given normalized eigenfunction, up to sign. On the finite collection of compact disks its strict signs have a positive absolute margin. Uniform convergence retains all those signs, possibly after a global sign change. If no sufficiently small finite-norm neighborhood retained them, a sequence contradicting this conclusion could be chosen and the same compactness argument would apply. Finally shrink the radius to put a closed finite-norm ball inside that neighborhood. □

6.2 Summable changes and separated exact eigenvalues

Proof of Theorem 1.1. Take monotone coordinate norms as in Section 2. Choose $b > 0$ such that the closed C^1 ball of radius b about $\mathcal{P}_0 = (g_0^{-1}, 1)$ consists of positive pairs inside $\mathcal{U}$. Set

$$l^{(j)} = \frac{1}{4} + 2^{-j-2} \quad (j \geq 0), \qquad \epsilon_0 = \gamma/4 > 0.$$

We construct $\mathcal{P}_j = (A_j, \rho_j)$, round for $t < l^{(j)}$, with a simple eigenvalue k_j^2 and certificate $\mathcal{C}_j$.

After stage i, apply Lemma 6.1 with interval $(k_i^2/2, 2k_i^2)$, and call its norm order and radius s_i, ε_i. At stage j, only earlier such numbers enter the requirements:

$$\begin{aligned} r_j &= \max\{j+1, 1, s_1, \dots, s_{j-1}\}, \\ \eta_j &= \min\left\{2^{-j},\, b2^{-j-1},\, \min_{1\le i<j} 2^{-(j-i+1)}\varepsilon_i\right\}. \end{aligned} \tag{6.3}$$

For $j = 1$ omit the inner minimum. Apply Proposition 2.2 to the fixed input $\mathcal{P}_{j-1}$, collar $(l^{(j)}, l^{(j-1)})$, order r_j and tolerance η_j. Its positive constant c_j is known before choosing k_j. Choose an allowed frequency beyond its threshold satisfying

$$k_j > 2k_{j-1} \quad (j > 1), \qquad c_j k_j^{\gamma/2} > j.$$

The resulting certificate then obeys

$$|\mathcal{C}_j| > j k_j^{1+\gamma/2}. \tag{6.4}$$

Use the disks constructed in Lemma 5.2; their centers are in $V \Subset \Omega$, and their radii are at most k_j^{-1}. Choose also the first frequency large enough to keep all cylinders in one fixed compact subregion of Ω. Only now select s_j, ε_j for this output. They restrict later stages and do not constrain the frequency that produced them.

For every fixed r, eventually $r_j \ge r$, and $\|\mathcal{P}_j - \mathcal{P}_{j-1}\|_{C^r} < 2^{-j}$. The sequence therefore converges in every C^r to a single smooth pair $\mathcal{P}_\infty = (A_\infty, \rho_\infty)$. Its total C^1 movement from the round pair is at most $b\sum_{j\ge1} 2^{-j-1} = b/2$, so it is positive and lies inside $\mathcal{U}$. For each i, (6.3) gives

$$\|\mathcal{P}_\infty - \mathcal{P}_i\|_{C^{s_i}} \le \sum_{j>i} 2^{-(j-i+1)}\varepsilon_i = \varepsilon_i/2.$$

Thus Lemma 6.1 applies simultaneously to every earlier certificate. There are nonzero real smooth v_i for the same limiting operator satisfying

$$-L_\infty v_i = \lambda_i v_i, \qquad k_i^2/2 < \lambda_i < 2k_i^2, \tag{6.5}$$

with certificate disks $\mathcal{C}_i$. The eigenvalue intervals are disjoint and tend to infinity because $k_{i+1} > 2k_i$.

An ordinary metric with the prescribed base operator

Put

$$G_\infty^{-1} = A_\infty/\rho_\infty, \qquad q = \frac{\rho_\infty \,\mathrm{d}\mu}{\mathrm{d}\,\mathrm{vol}_{G_\infty}}, \qquad g = G_\infty + q^2 \mathrm{d}\varphi^2 \quad \text{on } B \times S^1.$$

The ratio q of smooth positive densities is smooth and positive. Hence g is a smooth positive metric on a closed connected five-dimensional manifold, fixed independently of i. Its volume density is

$$\mathrm{d}\,\mathrm{vol}_g = q\,\mathrm{d}\,\mathrm{vol}_{G_\infty}\,\mathrm{d}\varphi = \rho_\infty\,\mathrm{d}\mu\mathrm{d}\varphi.$$

For any function f independent of φ, the coordinate divergence formula, with $\mu = m(x)\mathrm{d}x$, gives

$$\Delta_g f = \frac{1}{\rho_\infty m}\partial_a(\rho_\infty m G_\infty^{ab}\partial_b f) = \rho_\infty^{-1}\,\mathrm{div}_\mu(A_\infty \mathrm{d}f) = L_\infty f. \tag{6.6}$$

The pullbacks u_i of v_i are therefore exact real Laplace eigenfunctions on this single metric, at λ_i.

An intrinsic Hausdorff measure lower bound

Fix a closed interval I_0 of positive length in a circle coordinate chart. For one certificate cylinder $C_i^{(a)}$, let $D_i^{(a)}$ denote its transverse disk. Continuity and opposite endpoint signs force a zero on each joining segment. Consequently

$$\mathcal{Z}(u_i) \cap (C_i^{(a)} \times I_0)$$

projects onto $D_i^{(a)} \times I_0$ by the orthogonal base-coordinate projection and the circle coordinate.

All these cylinders lie in a single compact coordinate region. Extend its coordinate functions, and the circle coordinate on I_0, smoothly with cutoffs to the compact manifold. Their gradients have fixed bounds in g. Integrating along curves bounds their differences by a fixed constant times the intrinsic distance. Composing with any orthogonal projection gives one Lipschitz constant K, uniform in every disk's radius and orientation. The Hausdorff covering inequality therefore implies

$$\mathcal{H}_g^4(\mathcal{Z}(u_i) \cap (C_i^{(a)} \times I_0)) \geq K^{-4} \mathcal{H}_{\mathrm{Euc}}^4(D_i^{(a)} \times I_0) \geq c\, \mathcal{H}_{\mathrm{Euc}}^3(D_i^{(a)}) |I_0|. \tag{6.7}$$

The last constant is purely dimensional: the product disk has four-dimensional Lebesgue volume equal to the disk's three-volume times $|I_0|$, while a set of diameter d in its four-dimensional affine span has volume at most Cd^4. Apply this estimate to every countable cover and take the Hausdorff infimum. The identical three-dimensional covering normalization identifies the disk's three-volume with a fixed multiple of $\mathcal{H}_{\mathrm{Euc}}^3$. Thus no convention on the normalization of Hausdorff measure affects the bound.

The finitely many closed cylinders are in disjoint open balls, so their compact nodal subsets are pairwise positively separated. Hausdorff measure is additive on these separated sets directly from covers of sufficiently small diameter. Summing (6.7) and using (6.4) yields

$$\mathcal{H}_g^4(\mathcal{Z}(u_i)) \geq c|\mathcal{C}_i| > c\, i k_i^{1+\gamma/2}. \tag{6.8}$$

Only continuity and strict signs were used; no regularity of the nodal set is needed. Finally, (6.5) and $\epsilon_0 = \gamma/4$ give

$$\lambda_i^{1/2+\epsilon_0} \leq 2^{1/2+\epsilon_0} k_i^{1+\gamma/2}, \qquad \frac{\mathcal{H}_g^4(\mathcal{Z}(u_i))}{\lambda_i^{1/2+\epsilon_0}} \geq c\, 2^{-1/2-\epsilon_0} i \longrightarrow \infty.$$

The metric and positive exponent are fixed for the full sequence. This proves Theorem 1.1. □

Attribution and AI use. The counterexample, envelope construction, local waves, coefficient correction and iteration are due to OpenAI's preprint [1]. This note adapts that publicly released construction into an exposition of its implication for AIM Problem 011. Codex assisted with primary-source retrieval, proof reconstruction, mathematical checking, drafting and typesetting. These AI checks do not constitute independent human certification. No formal proof build is claimed, and no numerical experiment is used as evidence for the theorem.

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